Project Reflection

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Course : CS3081 - Cloud Computing
Project Title : Knowledge Distillation and Deployment of a Lightweight Command-Parsing Language Model on a Local Cloud.

Challenges Faced :

One of the biggest challenges was running the Phi-3.5 model locally. When I tried processing multiple requests at the same time, my system started running low on RAM. Because of this, I had to switch to sequencial processing to avoid system instability.

Another problem was that the teacher model sometimes generated incomplete or incorrectly formatted JSON responses. In many cases, the output was missing some required fields or did not follow the expected schema. I improved the retry process by providing the validation error as feedback to the model, so that it could correct the output in the next attempt. This helped to recover some of the previously failed examples.

I also faced difficulties while configuring Docker and the local object-storage setup. Initially, I had some issues in starting the container and setting the correct bind flags for SeaweedFS. After a few attempts, I was able to run the S3-compatible storage locally, I tested the integration by uploading and downloading the command dataset and verified the files using SHA-256 checksum.

How I Solved Them :-

I used checkpoints so that previously processed records would not need to be generated again after any interruption. I also added error tracking and validation to separate correct labels from failed outputs.
For object storage, I tested the integration using SeaweedFS 3.80. I uploaded and downloaded the command dataset and verified that both files had matching SHA-256 checksums. I also tested the worker's S3 download and upload functionality.
I ran the available unit tests and the NACL simulator to verify the implemented components.

Findings :-

This project helped me understand that running a machine learning model locally requires careful management of system resources. Increasing parallel requests does not always improve performance if the system has limited memory.
I also learned the importance of data validation, error handling, checkpoints and reproducibility. The project gave me practical exposure to how different cloud computing components work together.

Future Improvements :-

In the future, I would like to complete the full dataset labelling process, train the smaller student model and compare its output with the teacher model. I would also like to complete Kubernetes deployment and perform load testing to measure response time and throughput.

Overall, this project improved my understanding of cloud computing and practical machine learning workflows. It also taught me that debugging and handling unexpected errors are important parts of software development.